

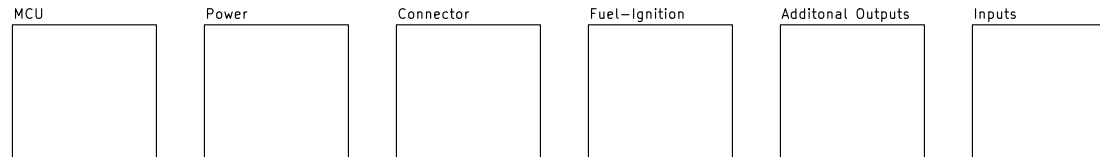
Pre-Ignition x4

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PN:MX4D

Pre-Ignition x4 is a 4 channel, (4 individually controlled ignition and injector outputs) Universal Standalone Engine Managment System.

Board Selection in tuner studio for this ECU is "Speeduino v0.4"



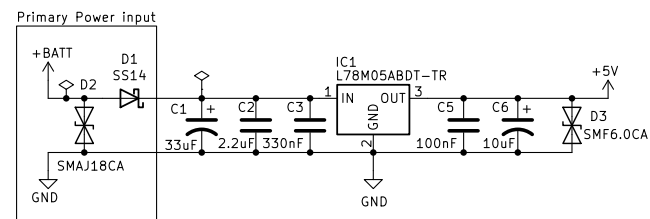
OpenLogicEFI ECUs are powered by an open source project called Speeduino by Josh Stewart.

OpenLogicEFI GitHub – <https://github.com/OpenLogicEFI>
Speeduino Official GitHub – <https://github.com/noisymime/speeduino>

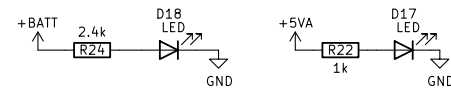
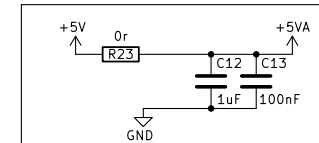
If you are happy with DetonationEMS and Speeduino Compatible devices, Firmware and/or Software. Please consider sponsoring Josh on github so he can advance his efforts.

<https://github.com/sponsors/noisymime>

Power Supply



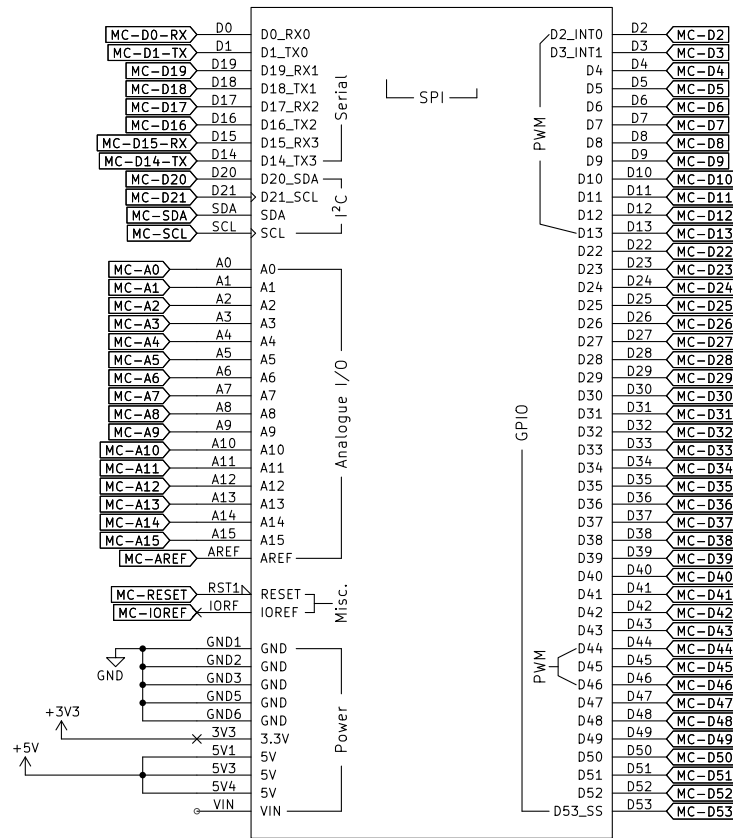
External 5v Reference.



Arduino Mega 2560 R3

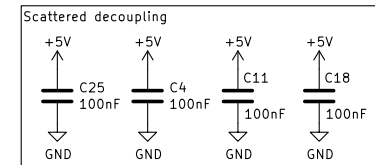
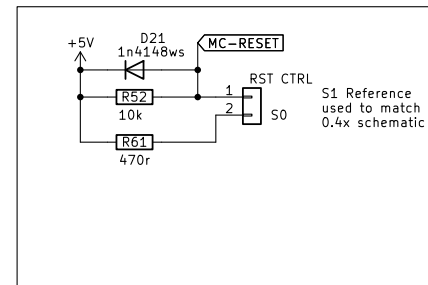
XA1

Arduino_Mega2560_Shield-Arduino



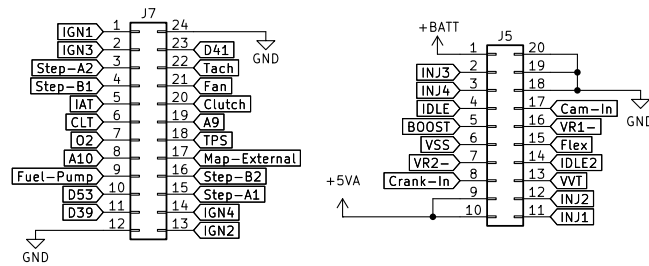
Remove DC Jack for proper fitment.

Reset Control

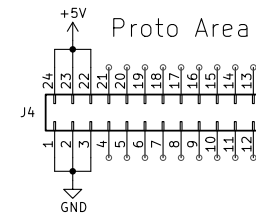
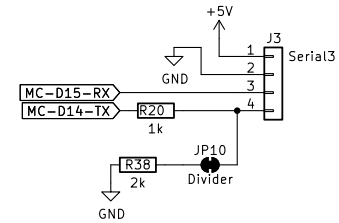
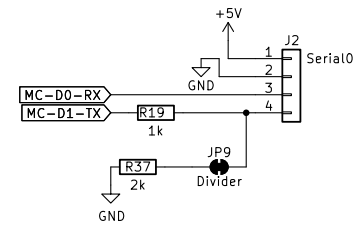


MCU

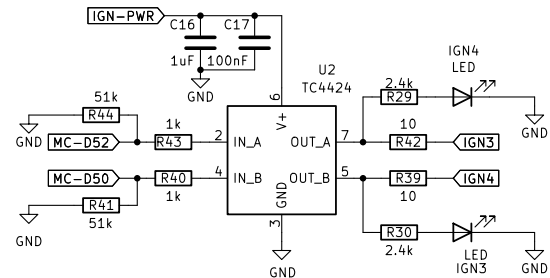
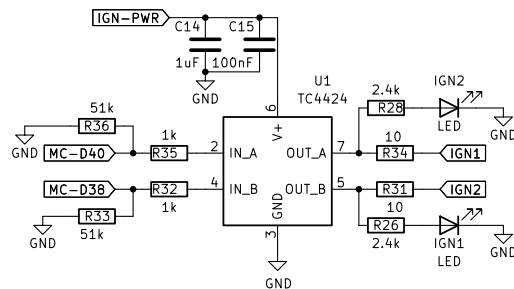
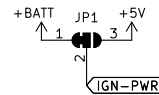
Main Connection



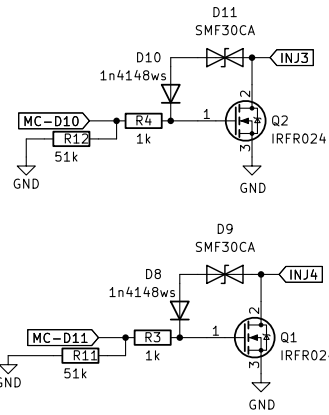
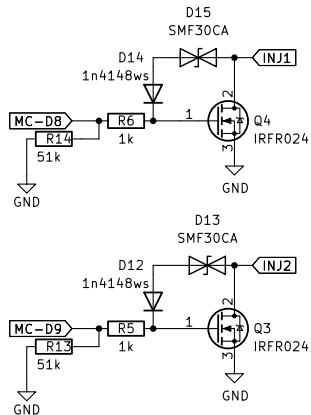
Serial/Proto



Ignition Output



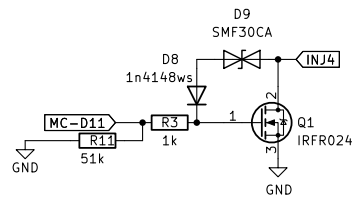
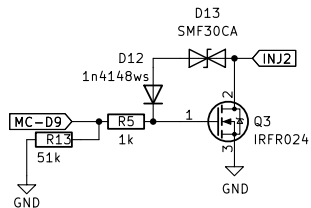
Fuel Injector Output



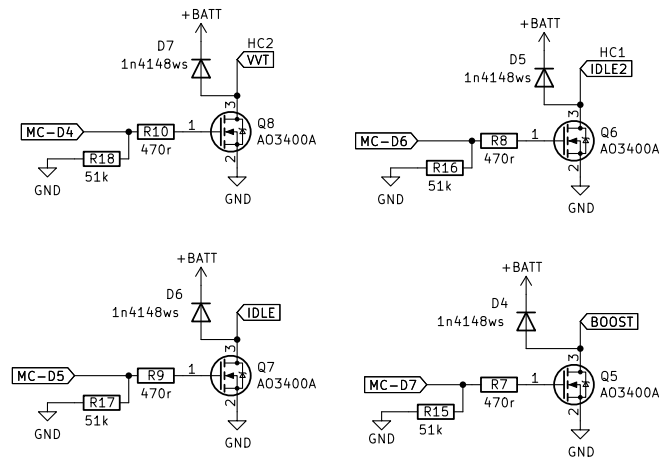
D6-D13 only need to be populated when NOT using internally clamped injector mosfets.

D6-D9 Fast switching

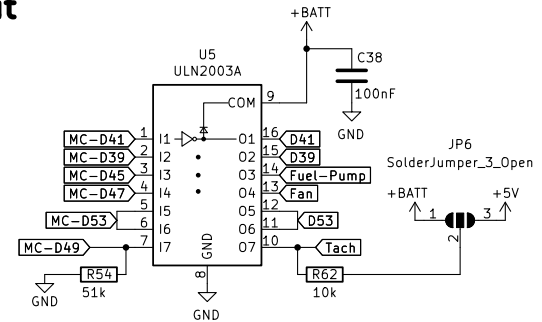
D10-D13 Zener Value must be below mosfet Vds. (Recommend Zener 36-60v)



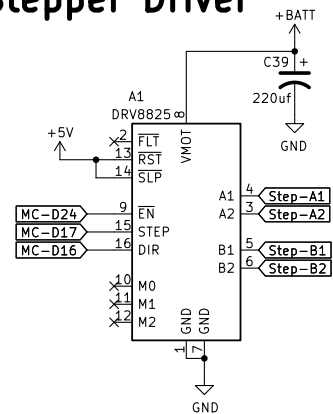
High Current Output – 1A Max



Relay Control Tach Output

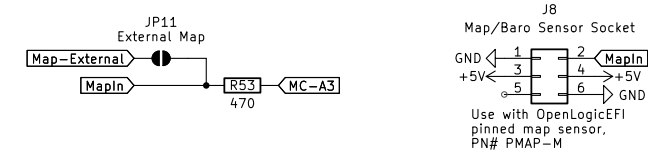


Stepper Driver

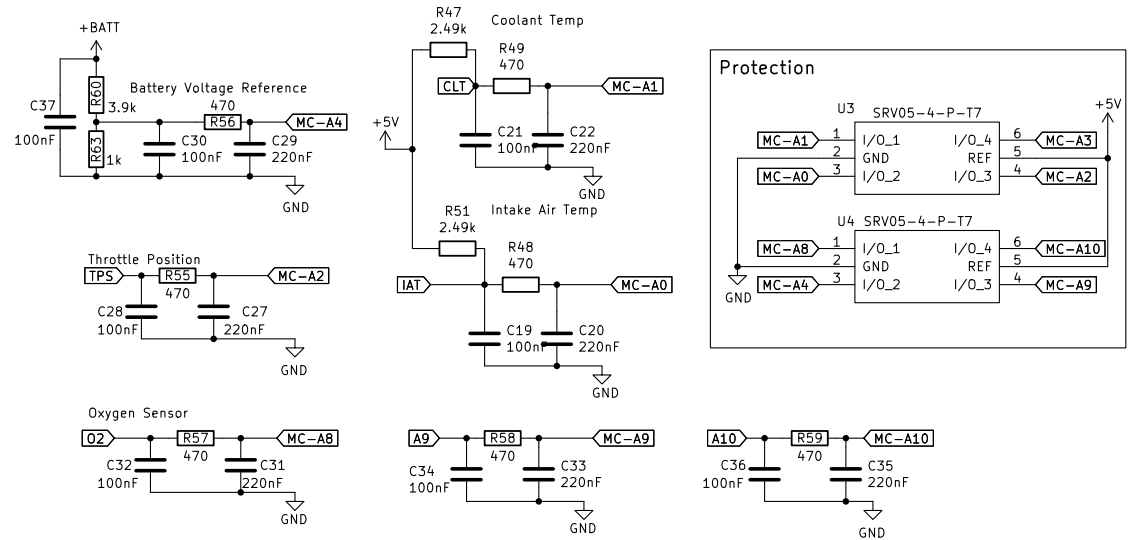
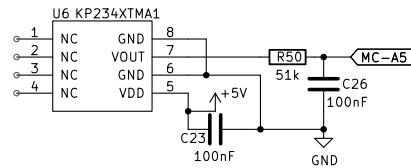


Analog-Inputs

Map/Baro Sensor



BARO Sensor



Digital-Inputs

Crank/Cam Inputs

